

free runtime with around 12Gb RAM, it provides access to its GPUs and TPUs that can accelerate the training process when working with huge datasets.

Colab has almost all the standard libraries pre-installed, offering users the plug-and-play functionality that saves them from the headache of library dependency. Moreover, the files generated while executing the codes can be safely saved in the Google drive itself by just mounting it and giving permission to Colab to access your drive. These advantages of Colab makes it very popular among the beginners, researchers, and developers to build their projects and access them on the go.

Work environment

This section can be skipped by beginners in Python. Work environment is completely online and can be visualised by visiting the link below in any Internet browser such as

tinyurl.com/EFYdiyWE, or
https://drive.google.com/drive/folders/1qI7pOwR_vlm7TF9p5jZ-xU4--G4jEK9t?usp=sharing

The environment consists of four files as shown in Figure 2.

1. **OnlineResponses:** Google sheet containing participant information (provided by participants through a Google form created for an event).
2. **font_playfair.ttf:** True Type Font (.ttf) file which decides the font style being used for printing text on the image.
3. **BlankCertificate.jpg:** Image file consisting of the certificate.
4. **AutomaticCertificateGeneration.ipynb:** Ipython Notebook file where the code is to be executed.

Procedure to run the project

Step 1: Access Google Colaboratory by visiting the link below using an Internet browser like Google Chrome, Internet Explorer, or Mozilla Firefox. The longer version of URL (uniform resource locator) is provided in case tinyurl is deprecated such as
https://colab.research.google.com/drive/1y_FMhXHHr0jnACirAc5VBTFowXL9iuYq?usp=sharing

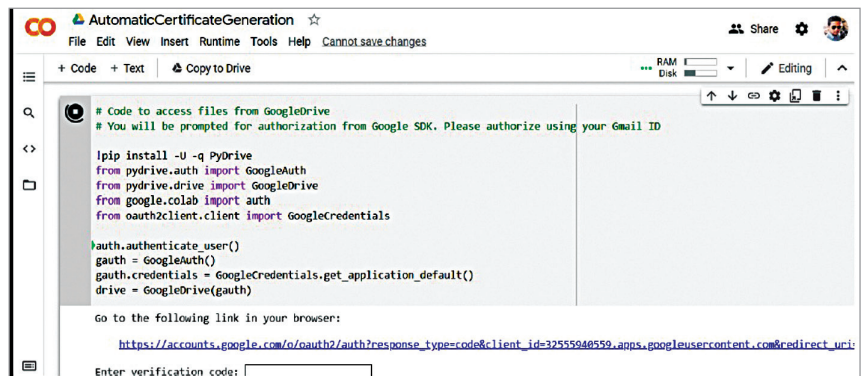


Figure 5: Authorising Colab to access drive

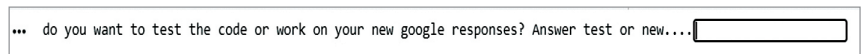


Figure 6: Testing the project



Figure 7: Successful generation of zipped file



Figure 8: Files generated and downloaded as a zipped folder

If not signed in, click on 'Sign in' button as shown in Figure 3 and join with your Google (Gmail) account.

Step 2: From the menu bar, select 'Runtime' and select 'Run All' as shown in Figure 4.

Step 3: Colab will ask for

authorisation to access your Google drive using 'Google Cloud SDK' as shown in Figure 5. Click on the link generated in the code and complete authorisation by pressing Enter. This is a security step and has to be completed every time this project is executed.